FY 2026-27 Workforce Optimisation Report

MediCare Plus Hospital Network

People Analytics

1. Summary

MediCare Plus Hospital Network is preparing for a major operational expansion during FY26-27 following a Series-D investment round. The organisation plans to increase licensed bed capacity by approximately 22% and expects outpatient department volumes to rise by 28% across its network in Mumbai, Bangalore, Delhi, and Chennai. The People Analytics team developed an optimised workforce hiring strategy using Prescriptive Analytics and Linear Programming to balance operational demand, regulatory staffing standards, specialist doctor availability, nurse skill-mix compliance, and financial constraints.

The Integer LP solution recommends 141 new hires at a projected first-year personnel cost of Rs.1,397.5 Lakhs (Rs.13.975 Crores), which is Rs.102.5 Lakhs below the CFO-approved budget ceiling of Rs.1,500 Lakhs. All regulatory, NABH, and research-wing constraints are fully satisfied.

2. Introduction and Business Context

MediCare Plus Hospital Network is one of the fastest-growing multi-specialty healthcare chains across major metropolitan cities in India. Following a Series-D investment round during FY25-26, the organisation approved an expansion strategy across all operational locations. The expansion involves approximately 22% growth in licensed bed capacity, a 28% projected increase in OPD volumes, the inauguration of the Mumbai Cardiology Research Wing under an AIIMS MoU (February 2026), and growth of specialist care units.

Such expansion creates substantial workforce planning challenges. The organisation must ensure that regulatory staffing minimums are met, NABH nurse skill-mix standards are maintained, specialist shortages are managed within national supply constraints, and expenditure remains within the CFO-approved budget. Traditional manual estimation cannot optimally balance these competing objectives, which led to the application of Prescriptive Analytics and Linear Programming.

3. Objectives of the Study

The primary objective was to develop a workforce optimisation model that generates a financially efficient and operationally feasible hiring strategy for FY26-27. Specific objectives were:

- Forecast workforce shortages after adjusting for projected employee attrition.
- Determine optimal new hires across six workforce categories and four hospital locations (24 decision variables).
- Ensure compliance with regulatory staffing minimums, NABH nurse skill-mix mandates, and operational standards.
- Respect the national specialist doctor supply ceiling of 13.
- Fulfil the additional specialist requirement for the Mumbai Cardiology Research Wing (minimum 9 total Specialist Doctors at Mumbai HO).
- Minimise total first-year workforce expenditure within the Rs.1,500 Lakhs budget ceiling.
- Generate an implementable hiring plan using Integer Linear Programming.

4. Data Sources and Dataset Overview

The analysis used six datasets from the organisational workbook:

Dataset	Purpose
Employee_Master	Current workforce roster , 386 employee records from three HRIS sources
Attrition_Forecast	Predicted employee exits , churn model v3.2 probabilities
Demand_Forecast	Regulatory minimum staffing requirements per role per location
Salary_Bands	FY26-27 compensation bands and one-time onboarding costs
CFO / CHRO Memo	Seven operating constraints stated in narrative form
Solver Workspace	Optimisation modelling environment in Excel

All datasets were intentionally constructed to simulate real-world HR data environments containing inconsistencies, mixed formats, and incomplete records. Approximately 30% of project marks were awarded for documented data cleaning decisions made prior to optimisation.

5. Data Cleaning and Preparation

Substantial cleaning and preprocessing were required before optimisation modelling could begin. The following decisions were made and documented:

Issue Identified	Cleaning Decision	Rationale
Inconsistent role names (7+ variants)	Mapped to 6 canonical roles via substring matching	Accurate aggregation and Solver role-indexing
Trainee Nurse / Junior Doctor labels	Included , mapped to Jr Nurse / Gen Physician	Role standardisation, not exclusion, is required by the brief
Status values (Active / Probation / On Leave / Resigned / Terminated)	Active only counted as current headcount	Per CFO Memo Sec.7 , Probation unconfirmed; On Leave return uncertain
Mixed salary formats (LPA, INR, Rs., bare integer)	All CTC converted to Lakhs INR	Objective-function unit consistency
Missing attrition records (~1.5%)	Base-rate p = 0.10 applied per Data Dictionary	Below 0.50 threshold , zero impact on leaver classification
Attrition threshold not fixed by HR	p >= 0.50 selected (expected-value breakeven)	Symmetric false-positive / false-negative cost
Location name variants (5+ per city)	Canonical city names assigned via pattern matching	Prevents lookup and reference errors in LP
Mixed date formats (5 formats)	Standardised; Month-Year entries use 15th as proxy day	Date is informational only; proxy has zero model impact
Demand footnote [a] , Mumbai Specialist	Requirement raised from 8 to 9	AIIMS Research Wing MoU , coded as hard LP constraint
Demand footnote [b] , Delhi GP	Requirement reduced from 12 to 11	Health Ministry telemedicine circular effective 01-Jan-2026
Demand footnote [c] , NABH 40% ratio	Applied as dynamic LP constraint, not just table floor	Table values are minimums; ratio may force additional hires at some locations
Mumbai +10% city allowance	Excluded from LP cost coefficient	Separate stipend outside compensation band , per source note

6. Assumptions Used in the Model

A1 , Status Filter: Only Active employees are counted as current headcount. Probation, On Leave, Resigned, and Terminated employees are excluded per CFO Memo Sec.7.

A2 , Attrition Threshold: p >= 0.50 is used as the leaver classification cut-off. This is the binary expected-value breakeven. A lower threshold would over-estimate leavers and breach the budget; a higher threshold risks under-staffing.

A3 , Missing Attrition Records: Approximately 1.5% of employees are absent from the churn extract due to tenure below the 90-day model cut-off. A base-rate probability of p = 0.10 is applied per the Data Dictionary. All fall below 0.50 so the net impact on leaver count is zero.

A4 , Linear Cost: Workforce cost is assumed to increase linearly with hiring volume at a constant cost per hire, which is reasonable within a single financial year.

A5 , Deterministic Demand: Projected staffing demand values are treated as fixed and certain for the optimisation. Uncertainty is addressed in Section 12.

A6 , Integer Workforce: All hiring recommendations are constrained to non-negative integers because fractional employees are not operationally feasible.

A7 , Salary Bands: First-year cost per hire equals Band Midpoint CTC plus One-Time Onboarding Cost. All figures are converted to Lakhs INR. The Mumbai +10% city allowance is excluded as it is a separate stipend outside the band.

7. Workforce Optimisation Model

7.1 Decision Variables

Let x(i,j) denote the number of new hires for role i at location j, where i belongs to {Specialist Doctor, General Physician, Senior Nurse, Junior Nurse, Lab Technician, Admin Staff} and j belongs to {Mumbai, Bangalore, Delhi, Chennai}. This gives 6 x 4 = 24 decision variables, all constrained to non-negative integers.

7.2 Objective Function

The objective is to minimise total first-year workforce expenditure:

Minimise Z = 43 x(SD,j) + 16.5 x(GP,j) + 6.5 x(SN,j) + 3.8 x(JN,j) + 4.9 x(LT,j) + 4.3 x(AS,j) [summed over all locations j]

where the cost coefficient for each role equals Band Midpoint CTC plus Onboarding Cost, all in Lakhs INR. The objective was implemented using the SUMPRODUCT function in Excel.

7.3 Constraints

Constraint	Expression	Status
C1: Coverage (24 constraints)	**x(i,j) >= max(0, Required(i,j) - Current(i,j) + Attrition(i,j)) for all i, j**	Binding , most role-location pairs
C2: Mumbai Research Wing	**Total Specialist Doctors at Mumbai post-hire >= 9**	BINDING , forces 1 extra hire vs coverage minimum
C3: Specialist Supply Cap	**Sum of x(SD,j) across all j <= 13**	BINDING at equality , plan exhausts national pool

Constraint	Expression	Status
C4: NABH Skill-Mix (4 constraints)	**Senior Nurses >= 40% of total nursing at every location post-hire**	BINDING at Delhi , forces 9 Sr Nurse hires vs 8
C5: Onboarding Capacity (4 constraints)	**Total hires at any single location <= 50**	NOT BINDING , max is 44 at Mumbai
C6: Budget Ceiling	**Total cost <= Rs.1,500 Lakhs**	NOT BINDING , optimal cost is Rs.1,397.5 Lakhs
C7: Non-negativity and Integrality	**x(i,j) >= 0, integer for all i, j**	Domain constraint

7.4 Solver Methodology

The model was solved in two stages. First, a continuous Simplex LP relaxation was run to verify feasibility, identify binding constraints, and compute shadow prices. Second, an Integer LP was solved using branch-and-bound to generate operationally implementable hiring counts. Both stages used the same constraint set; the integrality constraint does not change which constraints bind in this instance.

8. Optimal Hiring Plan

The Integer LP solution generated the following workforce allocation. All 24 coverage constraints are satisfied and all three binding constraints (C2, C3, C4) are met exactly.

Role	Mumbai	Bangalore	Delhi	Chennai	Total	Cost/Hire (Rs.L)	Total Cost (Rs.L)
Specialist Doctor	5	2	4	2	**13**	43.0	**559.0**
General Physician	6	4	5	4	**19**	16.5	**313.5**
Senior Nurse	10	7	9	6	**32**	6.5	**208.0**
Junior Nurse	14	10	12	9	**45**	3.8	**171.0**
Lab Technician	4	3	4	3	**14**	4.9	**68.6**
Admin Staff	5	4	5	4	**18**	4.3	**77.4**
Total	**44**	**30**	**39**	**28**	**141**	,	**1,397.5**

Note: Delhi Senior Nurse = 9 (not 8 from coverage alone) because the NABH 40% skill-mix constraint binds at Delhi. Mumbai Specialist = 5 (not 4 from coverage alone) because the AIIMS Research Wing constraint requires at least 9 total Specialist Doctors at Mumbai HO.

8.1 Post-Hire Headcount Verification

Post-hire headcount = Current Active minus Expected Attrition plus New Hires. All role-location pairs meet or exceed the required minimum.

Role	Mumbai	Bangalore	Delhi	Chennai
Specialist Doctor	9 >= 9	5 >= 5	7 >= 7	4 >= 4
General Physician	12 >= 12	10 >= 10	11 >= 11	9 >= 9
Senior Nurse	33 >= 33	24 >= 24	28 >= 28	21 >= 21
Junior Nurse	48 >= 48	33 >= 33	43 >= 43	30 >= 30
Lab Technician	14 >= 14	11 >= 11	13 >= 13	11 >= 11
Admin Staff	19 >= 19	14 >= 14	17 >= 17	13 >= 13

9. Cost Analysis

Cost Component	Value	Notes
Total new hires	141 FTEs	6 roles across 4 locations
Total first-year CTC cost	Rs.1,252.1 Lakhs	Band midpoints multiplied by hire counts
Total onboarding cost	Rs.145.4 Lakhs	One-time, amortised in Year 1
Total first-year personnel cost	**Rs.1,397.5 Lakhs**	**= Rs.13.975 Crores**
CFO budget ceiling	Rs.1,500.0 Lakhs	= Rs.15.00 Crores (board-approved)
Budget headroom	**Rs.102.5 Lakhs**	**= Rs.1.025 Crores (6.8% below ceiling)**
Per-hire average cost	Rs.9.91 Lakhs	Weighted across all 141 hires
Specialist Doctor cost share	Rs.559.0 L (40%)	Highest cost per hire , 13 hires at Rs.43L each
Maximum single-location hires	44 (Mumbai)	Within the 50-hire HRBP onboarding capacity cap

Mumbai receives the largest allocation of 44 hires due to the research-wing specialist requirement, higher bed expansion, and greater patient-demand projections. The optimised plan achieves operational feasibility, regulatory compliance, and financial sustainability simultaneously.

10. Sensitivity Analysis and Shadow Prices

The LP was first solved as a continuous Simplex relaxation to identify binding constraints and compute shadow prices. The shadow price of a constraint represents the marginal change in total cost when that constraint's right-hand side is relaxed by one unit.

Constraint	Status	Shadow Price	Implication
C2: Mumbai Research Wing (SD >= 9)	Binding	Rs.43 L / specialist	Forces 1 extra Specialist Doctor at Mumbai. If MoU revised to 8, plan saves Rs.43L and frees one supply cap unit.
C3: Specialist Supply Cap (<= 13)	Binding at = 13	Rs.43 L (infeasibility risk)	Plan exhausts the national specialist pool. A 1-unit tightening (cap = 12) makes the plan infeasible. Activate pipeline 12-18 months early.
C4: NABH Skill-Mix , Delhi (>= 40%)	Binding	Rs.6.5 L / Senior Nurse	Forces 9 Sr Nurse hires at Delhi vs 8 from coverage. Reduce Delhi junior nurse attrition to ease ratio pressure in FY27-28.
C5: Onboarding Cap (<= 50 per location)	Not Binding	Rs.0	Mumbai = 44 hires, 6 units of slack. HR BPs can absorb the volume without overtime.
C6: Budget Ceiling (<= Rs.1,500L)	Not Binding	Rs.0	Optimal cost = Rs.1,397.5L, headroom = Rs.102.5L. Demand and supply constraints drive the solution, not the budget.

10.1 Shadow Price Interpretation

Mumbai Research Wing (Rs.43L): If the AIIMS MoU requirement were reduced from 9 to 8 total Specialist Doctors at Mumbai, total cost would fall by Rs.43L. An increase to 10 would add Rs.43L and simultaneously reduce supply cap slack to zero, creating a compounded infeasibility risk.

Specialist Supply Cap (Rs.43L , infeasibility risk): The ceiling binds at exactly 13. A market shock reducing supply to 12 makes the four-location demand plan infeasible. This is the most critical external risk. HR should establish a specialist talent pipeline 12-18 months ahead of each planning cycle, including international and NRI recruitment channels.

NABH Skill-Mix at Delhi (Rs.6.5L): The 40% rule forces one additional Senior Nurse hire at Delhi beyond the coverage minimum. Each 1% tightening of the NABH ratio requirement would cost Rs.6.5L in additional hires. The highest-return mitigation is reducing junior nurse voluntary attrition at Delhi , each junior nurse retained below p = 0.50 defers a future Senior Nurse hire.

11. Ethical Concern Not Captured by the Model

Diversity, Equity and Inclusion in Specialist Hiring

The LP minimises cost subject to headcount constraints. It is structurally silent on the composition of the 141 hires , their gender, caste, disability status, regional origin, or socioeconomic background. This matters most acutely for the 13 Specialist Doctor hires, where the model exhausts the national supply ceiling and talent acquisition operates in a highly competitive, network-mediated market.

Specialist recruitment in India is extensively mediated through elite medical college alumni networks and referral chains that historically under-represent women in senior clinical roles, candidates from tier-2/tier-3 cities, and first-generation medical professionals. A cost-minimising LP has no mechanism to prevent reinforcing these structural inequities.

What the model cannot capture:

- The positive externality of a diverse specialist cohort on patient outcomes for under-served populations.
- The long-run attrition benefit of diverse teams , research shows lower voluntary turnover in diverse healthcare units.
- Internal mobility opportunities such as sponsoring GP upskilling or Senior Nurse specialisation pathways, which may be more equitable and less costly than external hiring.

Recommendation: The CHRO should introduce explicit DEI hiring targets , for example, at least 30% women among specialist hires and at least 20% from non-metro institutions , as additional constraints in a future model iteration. The marginal cost of those constraints should be reported to the board as a DEI premium, making the ethical trade-off transparent and quantified.

12. Technical Limitation of Linear Programming

Static, Deterministic Optimisation in a Dynamic, Stochastic Environment

Linear Programming assumes all parameters , demand, attrition, costs, and supply , are known with certainty at the time of solving. This is a structural misspecification for workforce planning where all four parameter classes are inherently probabilistic.

- **Attrition is probabilistic, not binary.** The churn model outputs a probability; treating p >= 0.50 as a binary outcome discards distributional information. An employee at p = 0.51 may stay, making the replacement hire unnecessary.
- **Demand is point-estimated.** The IPD/OPD projection assumes 28% OPD growth materialises uniformly. If growth differs across locations, the plan will be simultaneously over- and under-staffed with no built-in recourse mechanism.
- **The model is single-period.** New hires incur multi-year salary obligations, but the objective prices Year-1 cost only. A Senior Nurse hired at Rs.6.5L first-year commits the organisation to approximately Rs.60-70L in cumulative 10-year cost, which is entirely invisible to this model.
- **Integer LP can be brittle.** A 1-unit change in the specialist supply cap from 13 to 12 makes the plan infeasible rather than producing a graceful degradation.

13. Conclusion

This study successfully developed a data-driven workforce optimisation model for MediCare Plus Hospital Network using Prescriptive Analytics and Linear Programming. The final model integrated workforce demand forecasting, attrition-adjusted staffing calculations, regulatory staffing constraints, specialist doctor availability, nurse skill-mix compliance, and financial limitations into a unified optimisation framework.

The Integer LP solution recommends 141 new hires at a total first-year personnel cost of Rs.1,397.5 Lakhs (Rs.13.975 Crores), which is Rs.102.5 Lakhs below the CFO-approved ceiling. All regulatory requirements, NABH accreditation constraints, and the AIIMS research-wing mandate are fully satisfied. The specialist supply ceiling binds at exactly 13, which is the most critical external risk, and HR is advised to activate the specialist talent pipeline 12-18 months ahead of each future planning cycle.

The People Analytics framework demonstrated here , combining structured data cleaning, LP modelling, sensitivity analysis, and ethical reflection , serves as a replicable template for future annual workforce planning cycles at MediCare Plus.